

Integrated Hybrid Health Workforce Planning Model: A Comprehensive Framework

A Unified Approach Combining

1. Stock-and-flow supply projection
2. Needs-based demand estimation
3. Gap analysis
4. Optimization under constraints
5. Stochastic simulation
6. System dynamics modeling
7. Skill mix optimization

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1 Introduction

Health workforce planning is a critical component of health system strengthening, particularly for countries pursuing Universal Health Coverage (UHC) and Sustainable Development Goal 3 targets. Traditional workforce planning approaches often operate in silos, focusing either on supply projections or demand estimations without adequately integrating optimization, uncertainty analysis, and contextual factors. This comprehensive hybrid model synthesizes multiple methodological approaches into a unified framework that addresses these limitations.

The model integrates seven interconnected components: (1) **Stock-and-flow supply projection**, (2) **Needs-based demand estimation**, (3) **Gap analysis**, (4) **Optimization under constraints**, (5) **Stochastic simulation**, (6) **System dynamics modeling**, and (7) **Skill mix optimization**. This integration enables policymakers to move beyond simple workforce counts toward sophisticated planning that considers economic constraints, epidemiological trends, training pipeline dynamics, and equity considerations.

Relevant Citations: World Health Organization (2016) advocated for integrated workforce planning approaches in "Global Strategy on Human Resources for Health: Workforce 2030," while Liu et al. (2017) demonstrated the superiority of hybrid models in *Health Policy and Planning*.

2 Consolidated Mathematical Framework

2.1 Integrated Core Equations

The comprehensive model can be represented through three principal interconnected equations that capture supply, demand, and optimization simultaneously:

2.1.1 Supply-Demand-Optimization Integration

$$\text{Supply: } W_k(t+1) = W_k(t) + G_k(t) + R_k(t) + I_k(t) - [\alpha_k(t)W_k(t) + M_k^{out}(t) + D_k(t)] + M_k^{in}(t) \quad (1)$$

$$\text{Demand: } D_k(t) = \sum_{s \in S_k} \frac{P(t) \times p_s(t) \times \lambda_{s,k} \times C_s(t)}{\mu_{s,k}(t)} \times \left(1 + \delta \frac{\text{DALY}(t)}{\text{DALY}_{\text{global}}}\right) \quad (2)$$

$$\text{Objective: } \min \sum_{k=1}^K \sum_{r=1}^R [c_k W_{k,r}(t) + \phi_k \Delta_{k,r}(t)^2] \quad (3)$$

2.1.2 Dynamic Parameter Adjustments

Where dynamic parameters adjust based on context:

$$\alpha_k(t) = \alpha_{k,\text{base}} \times \left(1 + \beta \times \text{workload}_k(t) + \varepsilon \times \frac{\Delta_k(t-1)}{D_k(t-1)} \right) \quad (4)$$

$$\mu_{s,k}(t) = \mu_{s,k}^{\text{base}} \times \left(1 + \gamma \times \text{tech.change}(t) - \eta \times \text{burnout}_k(t) + \gamma_1 \times \frac{LE(t) - LE_{\text{ref}}}{LE_{\text{ref}}} \right) \quad (5)$$

$$B_r(t) = B_{\text{base}} \times \left(1 + \gamma_2 \times \frac{Y_r(t) - Y_{\text{ref}}}{Y_{\text{ref}}} \right) \quad (6)$$

2.2 Variable Definitions

Note: The model supports disaggregation by gender, rural/urban location, public/private sector, and facility level.

3 Comprehensive R Implementation

3.1 Main Simulation Function

```

1  # =====
2  # HYBRID HEALTH WORKFORCE PLANNING MODEL - R CODE
3  # =====
4
5  # Load required libraries
6  library(tidyverse)
7  library(deSolve)
8  library(lpSolve)
9  library(truncnorm)
10 library(ggplot2)
11 library(patchwork)
12
13 # =====
14 # 3.1 MAIN SIMULATION FUNCTION
15 # =====
16
17 run_hybrid_workforce_model <- function(params, n_years = 10,
18   n_simulations = 1000) {
19
20   # Initialize results storage
21   results <- list()
22
23   # =====
24   # 3.1.1 Stock-and-Flow Supply Projection
25   # =====
26
27   supply_model <- function(time, stocks, parms) {

```

```

27   with(as.list(c(stocks, parms)), {
28     # Dynamic attrition adjustment based on workload gap
29     workload_factor <- ifelse(D_total > 0, W_total / D_total, 1)
30     attrition_adj <- attrition_base * (1 + beta_attrition * (1 -
31       workload_factor))
32
33     # Dynamic productivity adjustment
34     productivity_adj <- productivity_base *
35       (1 + gamma_tech * tech_change - eta_burnout * (1 -
36         workload_factor))
37
38     # Flows
39     graduates <- min(training_capacity, training_seats * (1 +
40       0.1 * (D_total - W_total)/D_total))
41     attrition <- W_total * attrition_adj
42     net_migration <- migration_in * (1 + 0.2 * (1 -
43       workload_factor)) -
44       migration_out / (1 + 0.2 * (1 - workload_factor))
45
46     # Differential equations
47     dW_dt <- graduates - attrition + net_migration
48     dD_dt <- pop_growth * D_total * (1 + delta_daly *
49       (daly_rate/daly_global))
50
51     return(list(c(dW_dt, dD_dt),
52       attrition_rate = attrition_adj,
53       productivity = productivity_adj))
54   })
55 }
56
57 # Initial conditions
58 stocks <- c(W_total = params$initial_workforce,
59   D_total = params$initial_demand)
60
61 # Run supply-demand simulation
62 sim_results <- ode(y = stocks,
63   times = 0:n_years,
64   func = supply_model,
65   parms = params)
66
67 results$supply_demand <- as.data.frame(sim_results)
68
69 # =====
70 # 3.1.2 Multi-Cadre Disaggregation
71 # =====
72
73 simulate_cadres <- function(base_results, cadres, params) {
74   cadre_results <- list()
75
76   for (cadre in cadres) {
77     # Cadre-specific parameters
78     cadre_params <- params[[paste0("cadre_", cadre)]]
79
80     # Run cadre-specific model
81     cadre_sim <- ode(
82       y = c(W = cadre_params$initial, D =
83         cadre_params$demand_initial),
84       times = 0:n_years,

```

```

79     func = function(time, state, parms) {
80       with(as.list(c(state, parms)), {
81         # Cadre-specific dynamics
82         graduates <- parms$grad_rate * parms$training_capacity
83         attrition <- W * parms$attrition * (1 + 0.3 *
84           (parms$demand_initial - W)/parms$demand_initial)
85         migration_net <- parms$migration_in - parms$migration_out
86
87         dW <- graduates - attrition + migration_net
88         dD <- parms$demand_growth * D
89
90         return(list(c(dW, dD)))
91       })
92     },
93     parms = cadre_params
94   )
95
96   cadre_results[[cadre]] <- as.data.frame(cadre_sim)
97 }
98
99 return(cadre_results)
100 }
101
102 # Define cadres (example)
103 cadres <- c("physicians", "nurses", "midwives",
104   "clinical_officers", "chws")
105 results$cadre_specific <- simulate_cadres(results$supply_demand,
106   cadres, parms)
107
108 # =====
109 # 3.1.3 Optimization Module
110 # =====
111
112 optimize_allocation <- function(supply, demand, costs, budget,
113   constraints) {
114
115   n_regions <- length(unique(supply$region))
116   n_cadres <- length(unique(supply$cadre))
117
118   # Objective: Minimize weighted gaps and costs
119   f.obj <- rep(costs, each = n_regions)
120
121   # Constraint matrix
122   f.con <- matrix(0, nrow = 3 + n_regions + n_cadres,
123     ncol = n_regions * n_cadres)
124
125   row_idx <- 1
126
127   # Budget constraint
128   f.con[row_idx, ] <- rep(costs, each = n_regions)
129   constraint_dir <- c("<=")
130   constraint_rhs <- c(budget)
131   row_idx <- row_idx + 1
132
133   # Minimum coverage per region
134   for (r in 1:n_regions) {
135     f.con[row_idx, ((r-1)*n_cadres + 1):(r*n_cadres)] <- 1
136     constraint_dir <- c(constraint_dir, ">=")
137   }
138 }

```

```

133     constraint_rhs <- c(constraint_rhs,
134       constraints$min_coverage[r])
135   row_idx <- row_idx + 1
136 }
137 # Skill mix constraints
138 for (c in 1:(n_cadres-1)) {
139   for (r in 1:n_regions) {
140     col1 <- (r-1)*n_cadres + c
141     col2 <- (r-1)*n_cadres + c + 1
142     f.con[row_idx, col1] <- -constraints$skill_ratios[c]
143     f.con[row_idx, col2] <- 1
144     constraint_dir <- c(constraint_dir, ">=")
145     constraint_rhs <- c(constraint_rhs, 0)
146     row_idx <- row_idx + 1
147   }
148 }
149
150 # Solve
151 solution <- lp("min", f.obj, f.con, constraint_dir,
152   constraint_rhs,
153   all.int = TRUE)
154 return(solution)
155 }

```

Listing 1: Main Hybrid Workforce Planning Function

3.2 Parameter Definition

```

1  # =====
2  # 3.2 PARAMETER DEFINITION
3  # =====
4
5  model_parameters <- list(
6    # Core parameters
7    initial_workforce = 50000,
8    initial_demand = 55000,
9    n_years = 10,
10
11    # Stock-and-flow parameters
12    attrition_base = 0.05,
13    attrition_alpha = 5,      # Beta distribution shape1 for attrition
14    attrition_beta = 95,     # Beta distribution shape2 for attrition
15    beta_attrition = 0.3,    # Sensitivity of attrition to workload
16    training_capacity = 6000,
17    training_sd = 500,
18    training_seats = 5500,
19    migration_in = 800,
20    migration_out = 1200,
21
22    # Demand parameters
23    population = 5e7,
24    pop_growth = 0.025,
25    daly_rate = 0.35,
26    daly_global = 0.30,

```



```

27     delta_daly = 0.5,          # DALY adjustment coefficient
28
29     # Productivity parameters
30     productivity_base = 500,   # cases per worker per year
31     gamma_tech = 0.02,        # Annual productivity improvement from
                                tech
32     eta_burnout = 0.01,       # Productivity loss from burnout
33
34     # Economic parameters
35     base_budget = 1e9,
36     gdp_per_capita = 2000,
37     gini_index = 0.35,
38     life_expectancy = 65,
39
40     # Cadre-specific parameters (example for physicians)
41     cadre_physicians = list(
42     initial = 8000,
43     demand_initial = 10000,
44     grad_rate = 0.8,
45     training_capacity = 1000,
46     attrition = 0.04,
47     migration_in = 100,
48     migration_out = 150,
49     demand_growth = 0.03
50 )
51 )

```

Listing 2: Model Parameter Definitions

3.3 Stochastic Simulation Module

```

1     # =====
2     # 3.1.4 Stochastic Simulation Module
3     # =====
4
5     monte_carlo_simulation <- function(params, n_simulations) {
6
7         mc_results <- matrix(NA, nrow = n_years + 1, ncol =
                                n_simulations)
8         attrition_samples <- rbeta(n_simulations,
9                                   shape1 = params$attrition_alpha,
10                                  shape2 = params$attrition_beta)
11         graduation_samples <- rnorm(n_simulations,
12                                    mean = params$training_capacity,
13                                    sd = params$training_sd)
14
15         for (i in 1:n_simulations) {
16             # Sample parameters
17             sim_params <- params
18             sim_params$attrition_base <- attrition_samples[i]
19             sim_params$training_capacity <- max(0, graduation_samples[i])
20
21             # Run simulation with sampled parameters
22             sim_stocks <- c(W_total = params$initial_workforce,
23                             D_total = params$initial_demand)
24

```

```

25     sim_run <- ode(y = sim_stocks, times = 0:n_years,
26     func = supply_model, parms = sim_params)
27
28     mc_results[, i] <- sim_run[, "W_total"]
29   }
30
31   # Calculate statistics
32   summary_stats <- data.frame(
33     year = 0:n_years,
34     mean = apply(mc_results, 1, mean),
35     median = apply(mc_results, 1, median),
36     lower_95 = apply(mc_results, 1, quantile, probs = 0.025),
37     upper_95 = apply(mc_results, 1, quantile, probs = 0.975),
38     sd = apply(mc_results, 1, sd)
39   )
40
41   return(list(results = mc_results, summary = summary_stats))
42 }
43
44 results$stochastic <- monte_carlo_simulation(params,
      n_simulations)

```

Listing 3: Stochastic Simulation with Monte Carlo

3.4 Equity and Context Adjustment

```

1   # =====
2   # 3.1.5 Equity and Context Adjustment
3   # =====
4
5   adjust_for_equity <- function(workforce, population, gini, daly,
6     le, gdp) {
7
8     # Equity adjustment factors
9     rural_factor <- 1.3 # 30% higher need in rural areas
10    poverty_factor <- 1 + 0.5 * gini # Higher need in unequal
11      regions
12    disease_factor <- 1 + 0.8 * (daly / max(daly, na.rm = TRUE))
13
14    # Resource adjustment based on capacity to pay
15    budget_factor <- 1 + 0.3 * (gdp - mean(gdp)) / sd(gdp)
16
17    # Adjusted requirement
18    adjusted_need <- workforce * rural_factor * poverty_factor *
19      disease_factor
20    adjusted_budget <- params$base_budget * budget_factor
21
22    return(list(adjusted_need = adjusted_need,
23      adjusted_budget = adjusted_budget,
24      equity_factors = c(rural = rural_factor,
25        poverty = poverty_factor,
26        disease = disease_factor,
27        budget = budget_factor)))
28  }
29
30  # Apply equity adjustments

```

```

28 results$equity_adjusted <- adjust_for_equity(
29 workforce = results$supply_demand$W_total,
30 population = params$population,
31 gini = params$gini_index,
32 daly = params$daly_rate,
33 le = params$life_expectancy,
34 gdp = params$gdp_per_capita
35 )

```

Listing 4: Equity Adjustment Module

4 Model Comparisons and Target Setting

4.1 WHO Benchmark Comparison

```

1  # WHO benchmark comparison function
2  compare_to_benchmarks <- function(workforce, population, year) {
3
4      who_benchmarks <- data.frame(
5          cadre = c("physicians", "nurses_midwives",
6                    "total_health_workers"),
7          who_minimum = c(1/1000, 3/1000, 4.45/1000),
8          who_target = c(1.5/1000, 4.5/1000, 6.5/1000)
9      )
10
11     comparisons <- workforce %>%
12     group_by(cadre, region) %>%
13     summarise(
14         current_density = last(available) / population,
15         required_density = last(required) / population,
16         gap_density = required_density - current_density,
17         meets_minimum = current_density >=
18             who_benchmarks$who_minimum[match(cadre,
19             who_benchmarks$cadre)],
20         meets_target = current_density >=
21             who_benchmarks$who_target[match(cadre,
22             who_benchmarks$cadre)],
23         .groups = 'drop'
24     )
25
26     return(comparisons)
27 }

```

Listing 5: WHO Benchmark Comparison Function

4.2 UHC Service Coverage Index Integration

```

1  calculate_uhc_coverage <- function(workforce_metrics,
2      service_indicators) {
3
4      # Weighted index based on tracer indicators

```

```

4      uhc_index <- workforce_metrics %>%
5      mutate(
6        service_capacity_score = pmin(workforce_density / who_target,
7                                       1),
8        distribution_equity_score = 1 - gini(workforce_density),
9        financial_protection_score = 1 - (oop_expenditure /
10                                         total_health_expenditure)
11      ) %>%
12      summarise(
13        uhc_service_coverage_index = weighted.mean(
14          c(service_capacity_score, distribution_equity_score,
15            financial_protection_score),
16          w = c(0.5, 0.25, 0.25)
17        )
18      )
19      return(uhc_index)
20    }

```

Listing 6: UHC Coverage Index Calculation

4.3 SDG 3 Target Alignment

5 Model Selection and Comprehensive Integration

5.1 Recommended Comprehensive Model Structure

Based on the analysis, the most comprehensive approach integrates:

Primary Model: System Dynamics with Optimization Feedback

- Uses differential equations for natural dynamics
- Incorporates optimization for decision points
- Includes stochastic elements for uncertainty

Supporting Models:

1. **Monte Carlo simulation** for risk assessment
2. **Linear programming** for resource allocation
3. **Needs-based estimation** for target setting

5.2 Justification for Comprehensive Model Selection

1. **Dynamic Nature:** Health workforce systems exhibit delays, feedback loops, and nonlinearities best captured by system dynamics (Sterman, 2000).
2. **Policy Relevance:** Optimization components ensure recommendations are feasible within constraints.
3. **Uncertainty Management:** Stochastic elements address real-world variability in attrition, migration, and disease burden.
4. **Equity Integration:** The model explicitly incorporates Gini coefficients and rural/urban differentials.
5. **Scalability:** Modular design allows application at national, regional, or facility levels.

5.3 Key Strengths of the Integrated Approach

- **Policy Levers Explicitly Modeled:** Training capacity, retention incentives, task shifting, and recruitment can all be tested.
- **Multiple Time Horizons:** Short-term (1-2 years), medium-term (3-5 years), and long-term (10+ years) projections.
- **Scenario Testing:** "Best case," "worst case," and "policy intervention" scenarios.
- **Equity Monitoring:** Tracks distribution across geographic and socioeconomic strata.
- **Budget Integration:** Links workforce planning to fiscal space analysis.

6 Implementation Roadmap

6.1 Phased Implementation

6.2 Data Requirements by Source

6.3 Validation and Calibration

1. **Historical Validation:** Compare model projections against 5-10 years of historical data.
2. **Expert Review:** Engage health workforce experts to validate assumptions.

3. **Sensitivity Testing:** Identify most influential parameters.
4. **Cross-country Benchmarking:** Compare with similar countries' experiences.

7 Conclusion

This comprehensive hybrid health workforce planning model provides a robust, evidence-based framework for strategic health workforce planning. By integrating multiple methodological approaches and explicitly addressing equity, economic constraints, and uncertainty, it moves beyond simple projections to provide actionable insights for policymakers.

The model's modular design allows for phased implementation according to data availability and technical capacity, while its comprehensive nature ensures all critical aspects of workforce planning are addressed. Regular updating with new data and recalibration based on changing contexts will maintain its relevance and accuracy over time.

For implementation support, technical assistance, or customization for specific country contexts, the model code and documentation can be adapted to local needs while maintaining methodological rigor and international comparability.

8 References and Citations

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7. Campbell, J., Buchan, J., Cometto, G., et al. (2013). Human Resources for Health and Universal Health Coverage: Fostering Equity and Effective Coverage. *Bulletin of the World Health Organization*, 91(11), 853-863.

A Appendix A: Complete R Code Implementation

```

1  # =====
2  # VISUALIZATION FUNCTIONS
3  # =====
4
5  plot_workforce_results <- function(results) {
6
7      p1 <- ggplot(results$supply_demand, aes(time)) +
8      geom_line(aes(y = W_total, color = "Supply"), size = 1.2) +
9      geom_line(aes(y = D_total, color = "Demand"), size = 1.2) +
10     geom_ribbon(aes(ymin = W_total, ymax = D_total, fill = "Gap"),
11               alpha = 0.3) +
12     scale_color_manual(values = c("Supply" = "steelblue", "Demand"
13                                = "darkred")) +
14     scale_fill_manual(values = "grey50") +
15     labs(title = "Workforce Supply vs Demand",
16          x = "Year", y = "Number of Health Workers") +
17     theme_minimal()
18
19     # Stochastic plot
20     stochastic_data <- results$stochastic$summary
21     p2 <- ggplot(stochastic_data, aes(year, mean)) +
22     geom_ribbon(aes(ymin = lower_95, ymax = upper_95), alpha =
23               0.2, fill = "steelblue") +
24     geom_line(size = 1.2, color = "steelblue") +
25     labs(title = "Stochastic Projection with 95% CI",
26          x = "Year", y = "Health Workers") +
27     theme_minimal()
28
29     # Gap analysis
30     gap_data <- results$supply_demand %>%
31     mutate(gap = D_total - W_total,
32            gap_percent = gap / D_total * 100)
33
34     p3 <- ggplot(gap_data, aes(time, gap_percent)) +
35     geom_area(fill = "orange", alpha = 0.5) +
36     geom_line(size = 1.2, color = "darkorange") +
37     geom_hline(yintercept = 10, linetype = "dashed", color =
38               "red") +
39     annotate("text", x = max(gap_data$time), y = 12,
40             label = "Critical Shortage Threshold", hjust = 1) +
41     labs(title = "Workforce Gap Percentage",
42          x = "Year", y = "Gap (%)") +
43     theme_minimal()
44
45     return((p1 / p2) | p3)
46 }
47
48 # =====
49 # SENSITIVITY ANALYSIS FUNCTION

```

```

46 # =====
47
48 sensitivity_analysis <- function(params, variations) {
49
50   sensitivity_results <- list()
51
52   for (param_name in names(variations)) {
53     param_values <- variations[[param_name]]
54     param_results <- data.frame()
55
56     for (value in param_values) {
57       # Modify parameter
58       test_params <- params
59       test_params[[param_name]] <- value
60
61       # Run model
62       test_result <- run_hybrid_workforce_model(test_params,
63         n_years = 5, n_simulations = 100)
64
65       # Extract key outcome
66       final_gap <- tail(test_result$supply_demand$D_total -
67         test_result$supply_demand$W_total, 1)
68
69       param_results <- rbind(param_results,
70         data.frame(parameter = param_name,
71           value = value,
72           final_gap = final_gap))
73     }
74
75     sensitivity_results[[param_name]] <- param_results
76   }
77
78   return(sensitivity_results)
79 }
80
81 # =====
82 # POLICY SCENARIO TESTING
83 # =====
84
85 test_policy_scenarios <- function(base_params) {
86
87   scenarios <- list(
88     base = base_params,
89
90     scenario_1 = { # Increased training capacity
91       p <- base_params
92       p$training_capacity <- p$training_capacity * 1.5
93       p$training_seats <- p$training_seats * 1.5
94       p
95     },
96
97     scenario_2 = { # Improved retention
98       p <- base_params
99       p$attrition_base <- p$attrition_base * 0.7
100       p
101     },
102
103     scenario_3 = { # Task shifting

```



```

103     p <- base_params
104     p$productivity_base <- p$productivity_base * 1.2
105     p
106   },
107
108   scenario_4 = { # Combined intervention
109     p <- base_params
110     p$training_capacity <- p$training_capacity * 1.3
111     p$attrition_base <- p$attrition_base * 0.8
112     p$productivity_base <- p$productivity_base * 1.1
113     p
114   }
115 )
116
117 scenario_results <- list()
118 for (scenario_name in names(scenarios)) {
119   scenario_results[[scenario_name]] <-
120     run_hybrid_workforce_model(
121       scenarios[[scenario_name]], n_years = 10, n_simulations =
122         500)
123 }
124
125 return(scenario_results)
126 }
127
128 # =====
129 # EXECUTION AND OUTPUT
130 # =====
131
132 # Run the comprehensive model
133 final_results <- run_hybrid_workforce_model(
134   model_parameters,
135   n_years = 10,
136   n_simulations = 1000
137 )
138
139 # Generate visualizations
140 workforce_plots <- plot_workforce_results(final_results)
141 print(workforce_plots)
142
143 # Perform sensitivity analysis
144 sensitivity_variations <- list(
145   attrition_base = seq(0.03, 0.08, by = 0.01),
146   training_capacity = seq(4000, 8000, by = 1000),
147   pop_growth = seq(0.015, 0.035, by = 0.005)
148 )
149
150 sensitivity_results <- sensitivity_analysis(model_parameters,
151   sensitivity_variations)
152
153 # Test policy scenarios
154 policy_scenarios <- test_policy_scenarios(model_parameters)
155
156 # =====
157 # REPORT GENERATION
158 # =====
159
160 generate_policy_report <- function(results, scenarios) {

```

```

158
159     report <- list()
160
161     # Key metrics
162     final_year <- tail(results$supply_demand$time, 1)
163     final_supply <- tail(results$supply_demand$W_total, 1)
164     final_demand <- tail(results$supply_demand$D_total, 1)
165     final_gap <- final_demand - final_supply
166     gap_percentage <- (final_gap / final_demand) * 100
167
168     report$key_metrics <- data.frame(
169       metric = c("Final Supply", "Final Demand", "Absolute Gap",
170                 "Gap Percentage"),
171       value = c(round(final_supply), round(final_demand),
172                round(final_gap), round(gap_percentage, 1))
173     )
174
175     # Risk assessment
176     stochastic_summary <- results$stochastic$summary
177     risk_level <- ifelse(mean(stochastic_summary$lower_95) <
178                          final_demand * 0.9,
179                          "High Risk", "Moderate Risk")
180
181     report$risk_assessment <- list(
182       risk_level = risk_level,
183       confidence_interval_width = mean(stochastic_summary$upper_95 -
184                                        stochastic_summary$lower_95),
185       probability_shortage =
186         mean(results$stochastic$results[nrow(results$stochastic$results),
187                                         ] < final_demand)
188     )
189
190     # Policy recommendations
191     recommendations <- c()
192
193     if (gap_percentage > 15) {
194       recommendations <- c(recommendations,
195                            "CRITICAL: Immediate expansion of training capacity
196                            required",
197                            "Implement emergency retention incentives")
198     } else if (gap_percentage > 10) {
199       recommendations <- c(recommendations,
200                            "HIGH PRIORITY: Scale up training by 20-30%",
201                            "Enhance retention programs")
202     } else if (gap_percentage > 5) {
203       recommendations <- c(recommendations,
204                            "MODERATE: Consider gradual training expansion",
205                            "Optimize skill mix through task shifting")
206     } else {
207       recommendations <- c(recommendations,
208                            "STABLE: Maintain current training levels",
209                            "Focus on quality improvement and distribution")
210     }
211
212     report$recommendations <- recommendations
213
214     # Scenario comparison
215     scenario_comparison <- data.frame()

```

```

210   for (scenario in names(scenarios)) {
211     scenario_gap <-
212       tail(scenarios[[scenario]]$supply_demand$D_total -
213         scenarios[[scenario]]$supply_demand$W_total, 1)
214     scenario_comparison <- rbind(scenario_comparison,
215       data.frame(scenario = scenario,
216         final_gap = scenario_gap))
217   }
218
219   report$scenario_analysis <- scenario_comparison
220
221   return(report)
222 }
223
224 # Generate final report
225 policy_report <- generate_policy_report(final_results,
  policy_scenarios)
  print(policy_report)

```

Listing 7: Complete Model Implementation

Mathematical Summary of Key Equations

Supply: $W_k(t+1) = W_k(t) + G_k(t) + R_k(t) + I_k(t) - [\alpha_k(t)W_k(t) + M_k^{out}(t) + D_k(t)] + M_k^{in}(t)$ (7)

Demand: $D_k(t) = \sum_{s \in S_k} \frac{P(t) \times p_s(t) \times \lambda_{s,k} \times C_s(t)}{\mu_{s,k}(t)} \times \left(1 + \delta \frac{DALY(t)}{DALY_{global}}\right)$ (8)

Objective: $\min \sum_{k=1}^K \sum_{r=1}^R [c_k W_{k,r}(t) + \phi_k \Delta_{k,r}(t)^2]$ (9)

Where:

- $\alpha_k(t) = \alpha_{k,base} \times \left(1 + \beta \times \text{workload}_k(t) + \varepsilon \times \frac{\Delta_k(t-1)}{D_k(t-1)}\right)$
- $\mu_{s,k}(t) = \mu_{s,k}^{base} \times \left(1 + \gamma \times \text{tech.change}(t) - \eta \times \text{burnout}_k(t) + \gamma_1 \times \frac{LE(t) - LE_{ref}}{LE_{ref}}\right)$
- $B_r(t) = B_{base} \times \left(1 + \gamma_2 \times \frac{Y_r(t) - Y_{ref}}{Y_{ref}}\right)$

Table 1: Variable Definitions and Data Sources

Symbol	Definition	Typical Data Sources
$W_k(t)$	Workforce of cadre k at time t	HRIS, professional registers
$D_k(t)$	Required workforce of cadre k at time t	Epidemiological surveys, service targets
$P(t)$	Population size	Census, population projections
$p_s(t)$	Prevalence of condition requiring service s	Disease registries, household surveys
$\lambda_{s,k}$	Service frequency per case for cadre k	Service statistics, time-motion studies
$C_s(t)$	Coverage target for service s (0-1)	National health strategies, UHC targets
$\mu_{s,k}(t)$	Productivity (services/worker/year)	Workforce productivity studies
$\alpha_k(t)$	Attrition rate for cadre k	Exit interviews, employment records
$G_k(t)$	New graduates	Education institution records
$B_r(t)$	Available budget in region r	Ministry of Finance, health accounts
$Y_r(t)$	Income measure (GDP/GNI per capita)	National statistics, World Bank
$LE(t)$	Life expectancy	Vital registration, DHS
$DALY(t)$	Disability-adjusted life years	Global Burden of Disease study

Table 2: SDG 3 Target Alignment in the Model

SDG Indicator	Model Metric	Target	Status
3.8.1 UHC service coverage	Health worker density	$\geq 4.45/1000$	Calculated
3.c Health workforce	Training rate	Match demand	Simulated
3.b Health financing	Budget allocation	$\geq 5\%$ GDP	Compared
10.2 Social inclusion	Gini coefficient for distribution	< 0.3	Monitored

Table 3: Data Requirements for Model Implementation

Data Category	Required Elements	Priority
HRIS Data	Current workforce by cadre, location, facility	High
Training Pipeline	Enrollment, graduation, attrition rates	High
Population Data	Census, projections, age structure	High
Epidemiological	Disease burden, service utilization	Medium
Economic	Health budget, salaries, GDP	Medium
Equity Indicators	Gini coefficient, rural/urban breakdown	Medium